



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE TITLE & CODE: DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

Title of the project	Survey Data Analysis Tool	
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Abstract

The Survey Data Analysis Tool is a data science project developed to simplify the process of analyzing survey data and converting it into meaningful, easy-to-understand insights. Surveys are widely used across businesses, educational institutions, healthcare, and research to collect opinions and information from large groups of people. However, raw survey data is often unorganized, contains missing or inconsistent responses, and is difficult to interpret directly, making it hard for decision-makers to draw useful conclusions without technical expertise.

This project addresses that problem by building a structured pipeline that takes a survey dataset — such as the Young People Survey, containing responses on music preferences, movie tastes, hobbies, phobias, and demographics — and processes it through multiple stages. The data is first cleaned by handling missing values, duplicates, and inconsistent entries. It is then explored using summary statistics to understand response patterns, followed by visualization through bar charts, pie charts, and histograms that make trends easy to see at a glance. Statistical testing, specifically the chi-square test, is applied to determine whether relationships between two survey variables, such as gender and movie preference, are statistically significant or occur by chance.

Finally, all cleaned data, visualizations, and statistical results are integrated into a single interactive dashboard, allowing users without technical or statistical background to explore findings and make informed decisions. Although demonstrated using the Young People Survey dataset, the tool is designed to work with any structured survey data, making it a flexible and reusable solution applicable to market research, employee feedback, academic evaluations, and public health surveys.

Justification of the Project

Survey data is one of the most common forms of data collected across nearly every industry — businesses, colleges, hospitals, and government bodies all rely on surveys to understand people's opinions, habits, and needs. However, the value of a survey lies not in collecting responses, but in correctly interpreting them. In practice, most survey data is left underutilized because analyzing it manually requires time, statistical knowledge, and technical tools that not everyone has access to. This creates a gap between data collection and data-driven decision-making — organizations gather large volumes of responses but often fail to extract real insights from them in a timely or accurate manner.